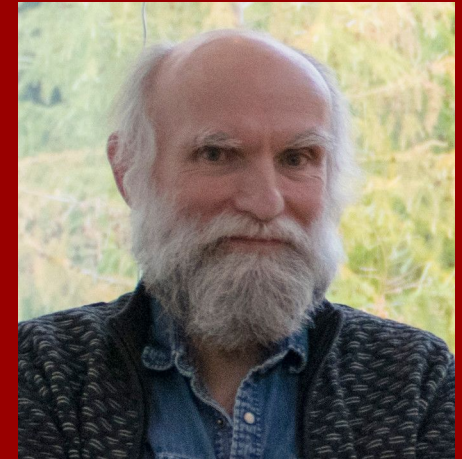


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Sampling effects

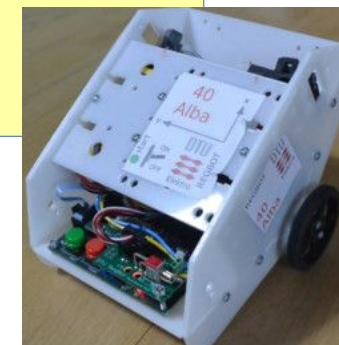
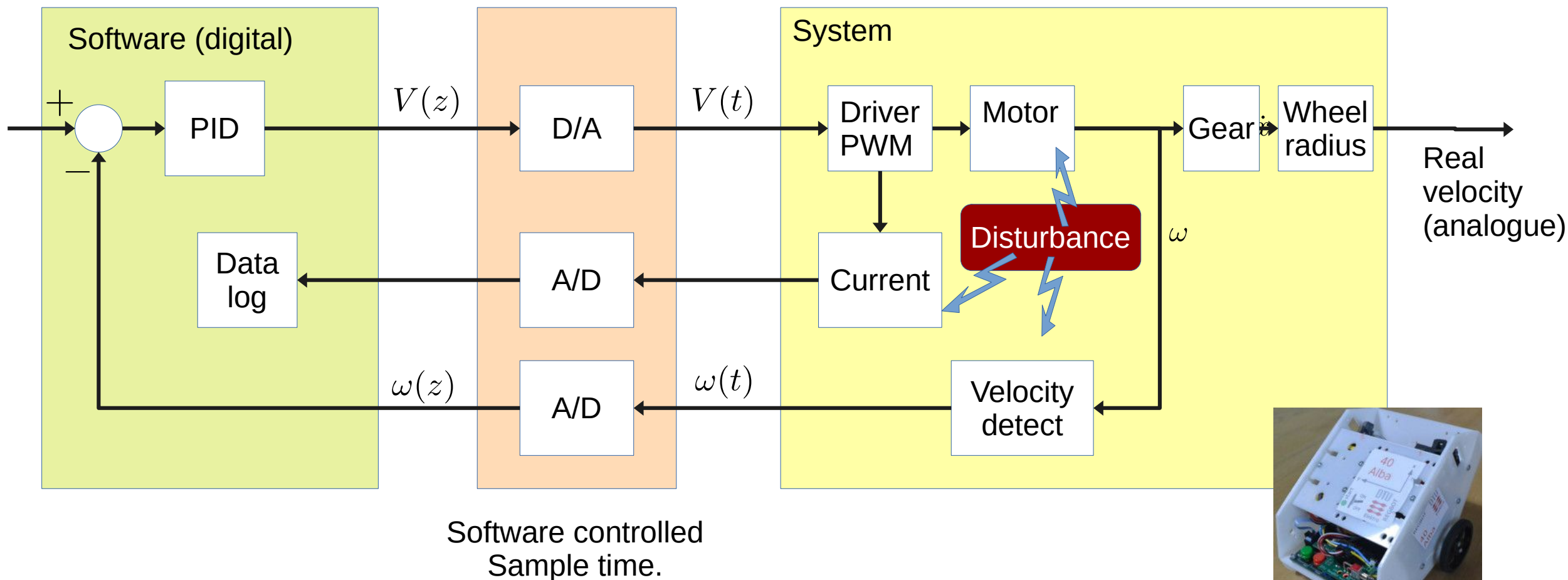
- Plan
- Sampling effect
 - Disturbances
 - Exersize on robot
 - Summary
- Minimum sampling frequency
 - P-controller in Matlab
 - ZOH effect
 - Sampling time effect
 - Exersize MATLAB
 - Exersize on Robot
 - Summary (next week)



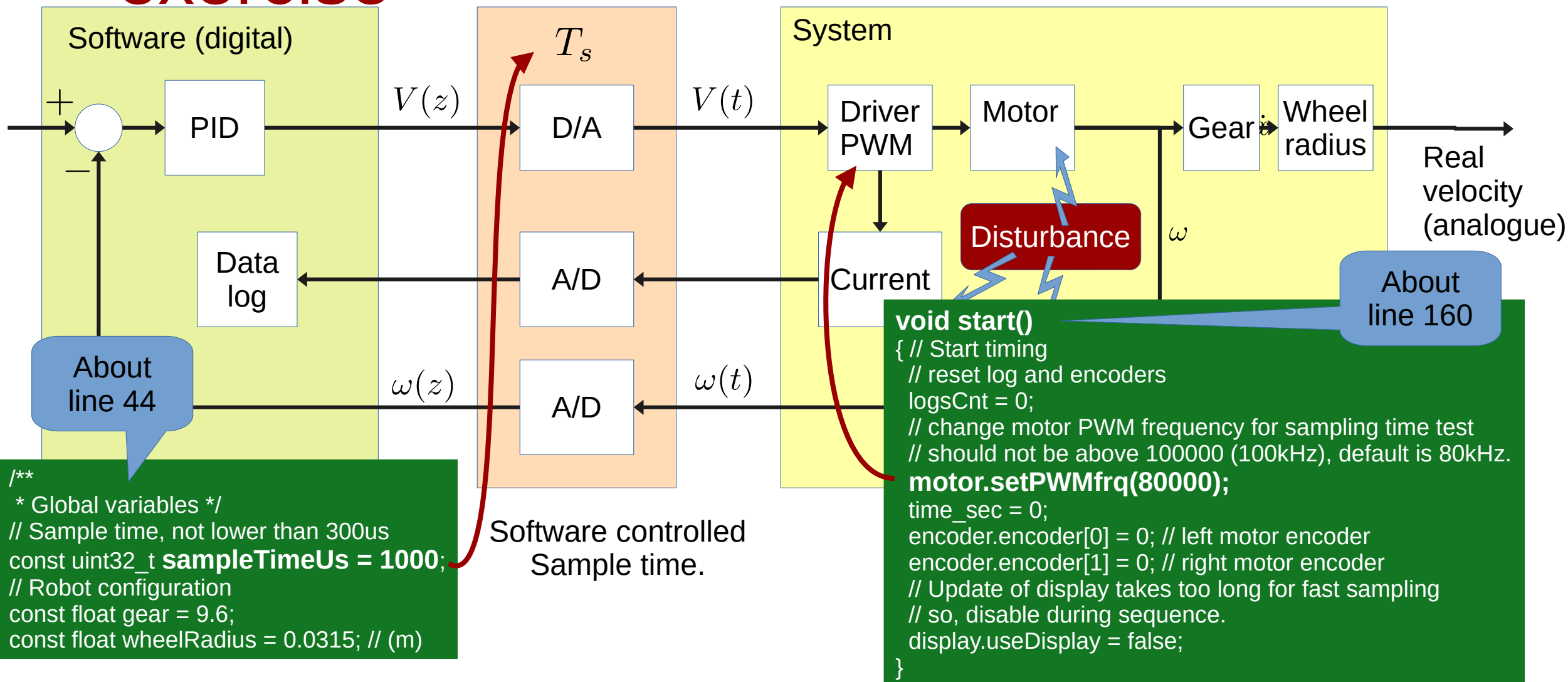
Christian Andersen

Week 5

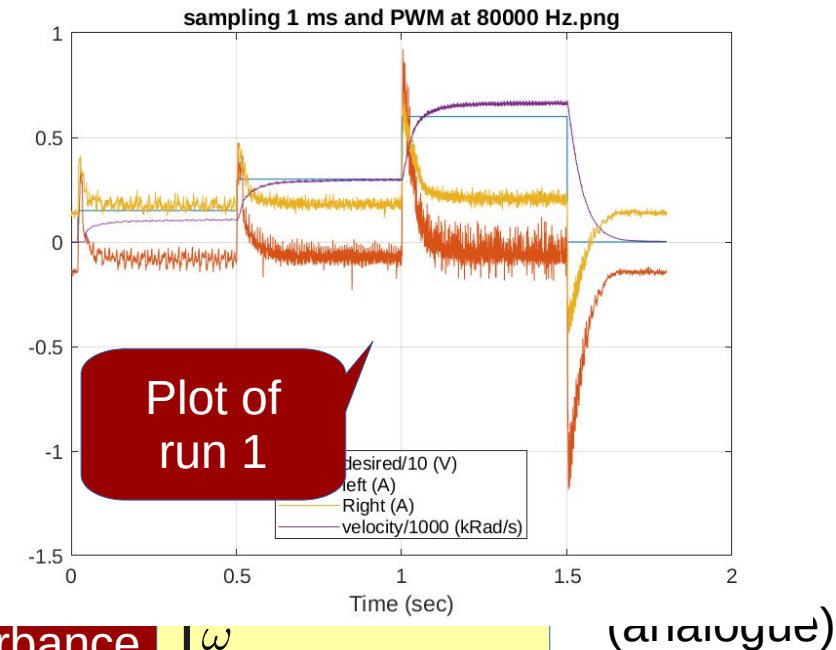
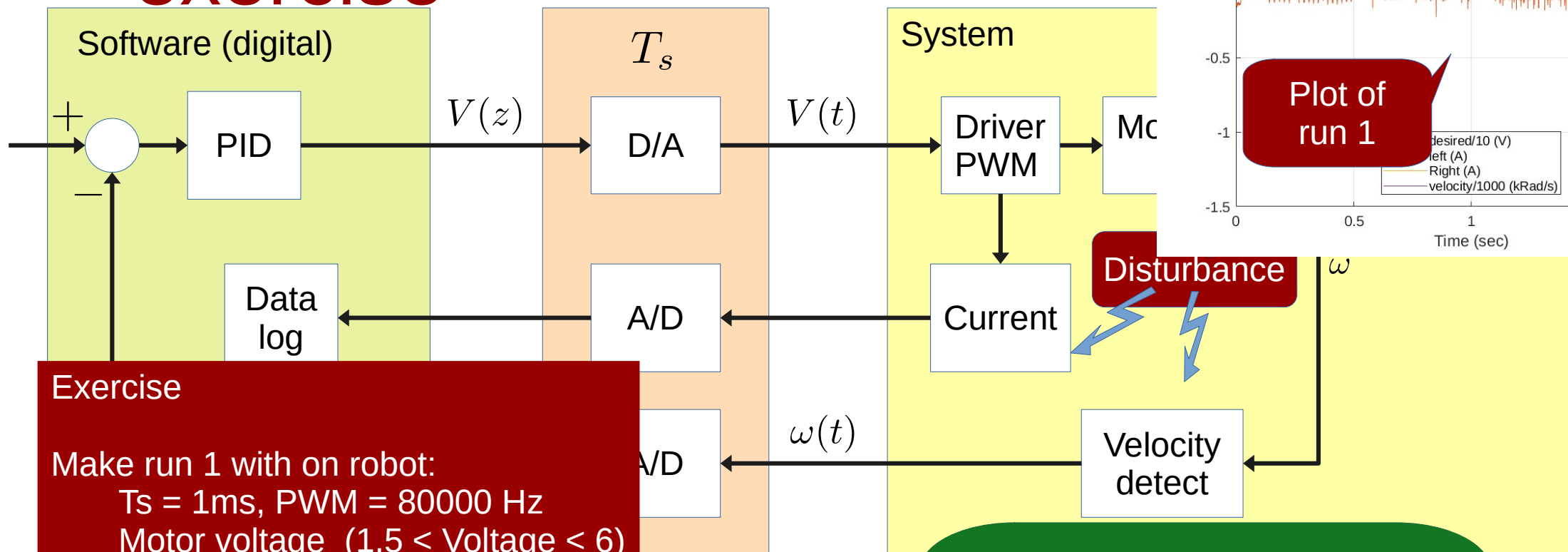
Basebot sampling effects



Basebot sampling effects exercise



Basebot sampling effects exercise



Exercise

Make run 1 with on robot:

$T_s = 1\text{ms}$, PWM = 80000 Hz

Motor voltage ($1.5 < \text{Voltage} < 6$)

Transfer data log to MATLAB,
plot: current (left and right)

Make run 2 with:

$T_s = 1\text{ms}$, PWM = 1990 Hz

Make the same plot

are controlled
sample time.

Observe the difference between the plots.

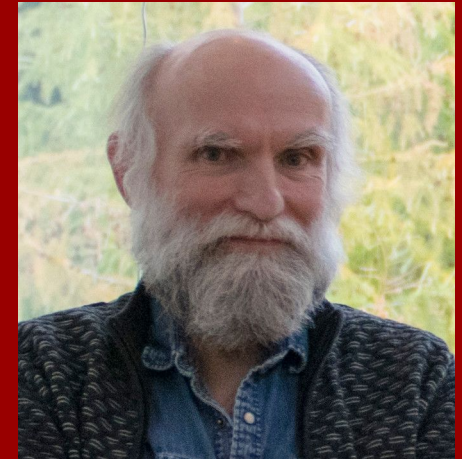
1. Dominating frequency of disturbance?

2. Do this match the theory?

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Sampling effects

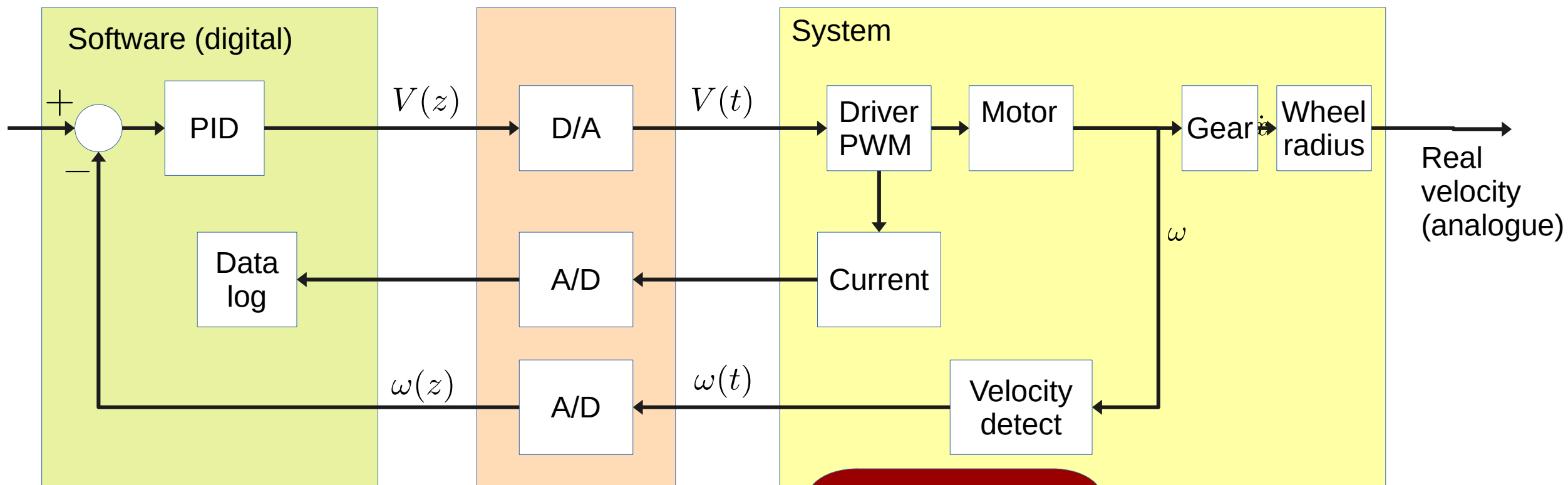
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Christian Andersen

Week 5

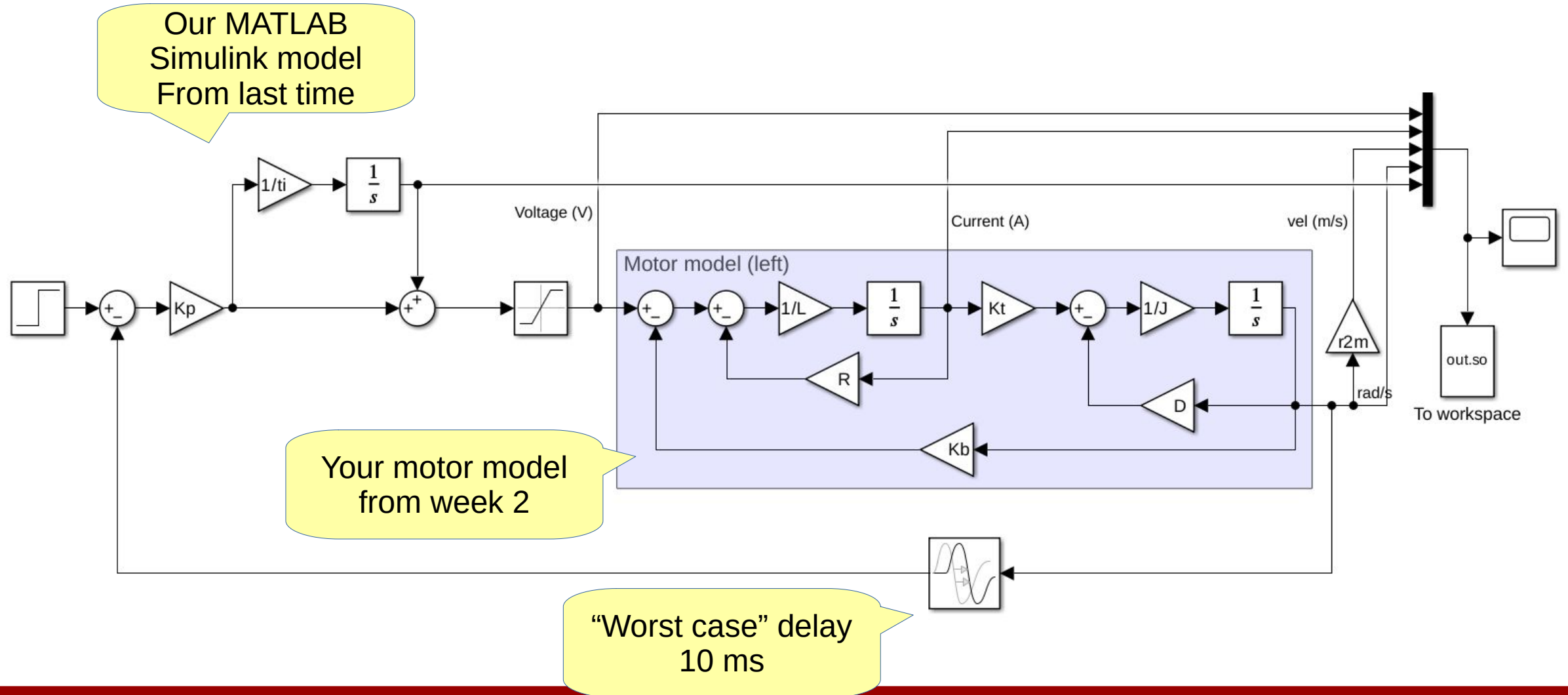
Sampling



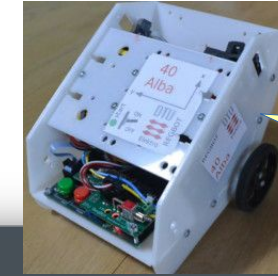
Software controlled
Sample time.

What is a good
sample time?

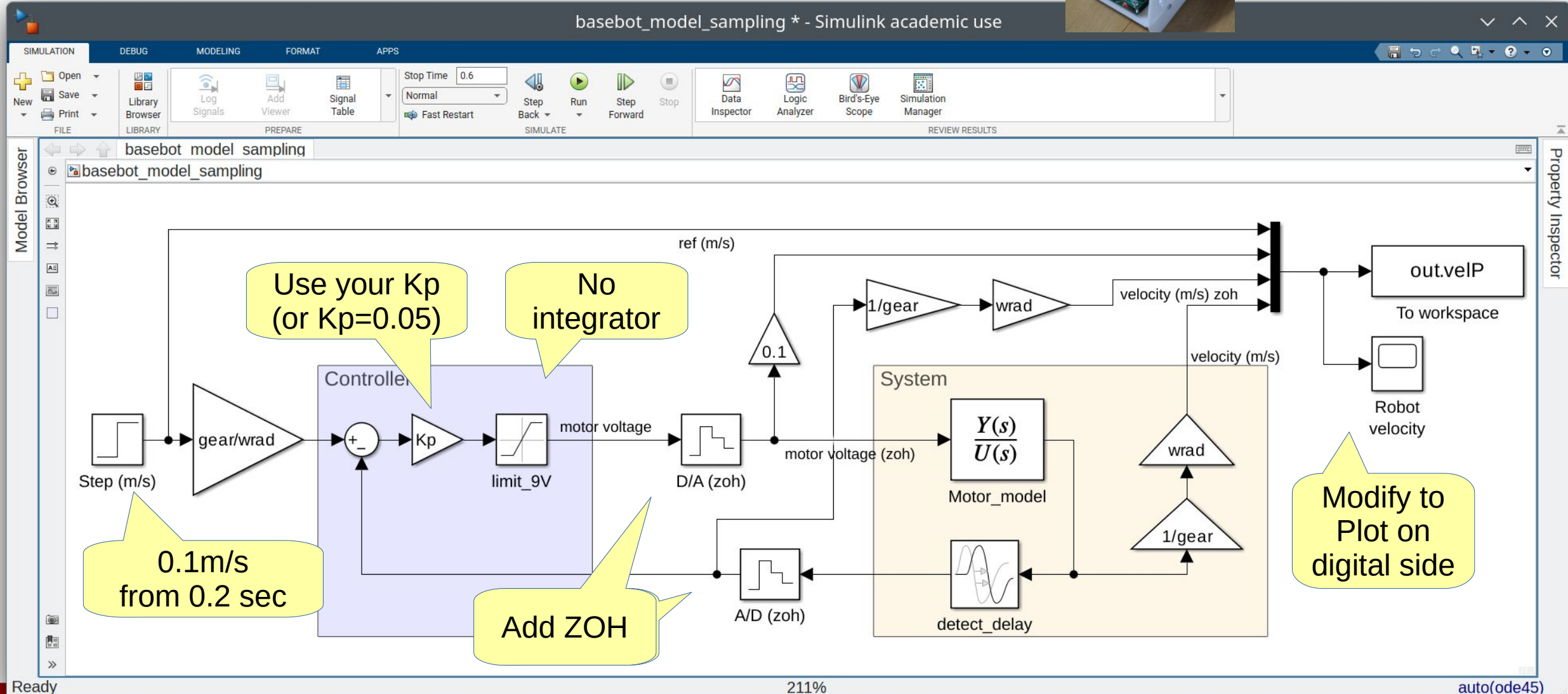
MATLAB Simulink with controller



Sampling frequency

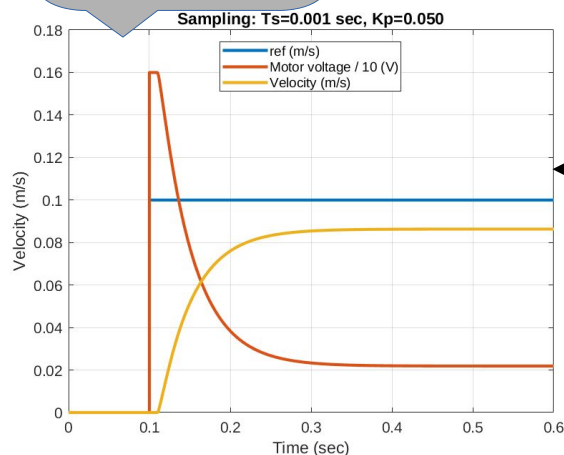


Gear: 9.6:1
Wheel radius 3 cm
(wheels up inertia)



Sampling frequency exercise (MATLAB and Robot)

There will be stationary error, but OK for now.



Modify your MATLAB Simulink model to:

- 1) Include D/A and A/D (ZOH) blocks
- 2) Make a variable for sample time (T_s), use it in ZOH blocks.
- 3) Robot mass not included in J.
- 4) Run and see you get a (good) plot

Experiment with longer sampling time

- 5) Until the system gets unstable.
- 6) Find the longest sample time with decent result.

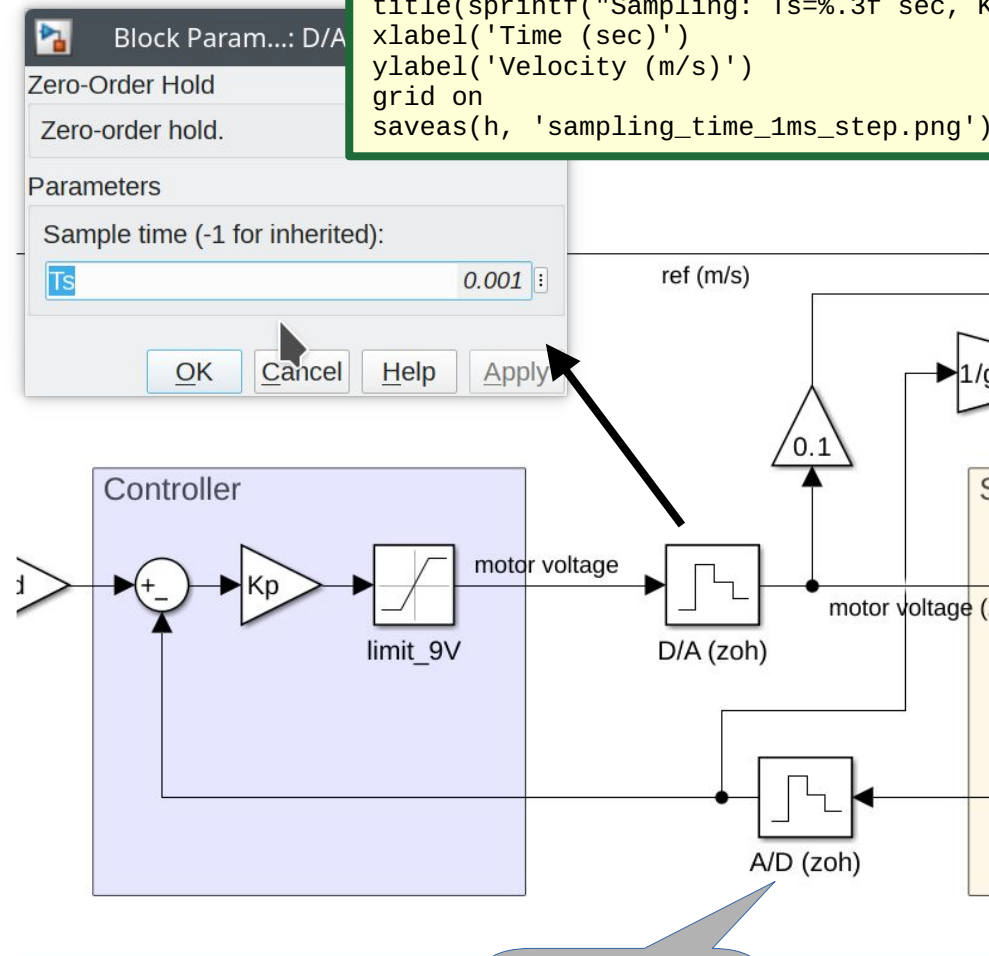
```
void controlUpdate()
{ // Measured wheel velocity (positive is forward)
  // float velLeft = -encoder.motorVelocity[0];
  // float velRight = encoder.motorVelocity[1];
  //
  // Desired velocity
  float velRef = desiredValue;
  //
  // Left motor voltage (-9V to +9V)
  motor.motorVoltage[0] = -velRef;
  // Right motor voltage (-9V to +9V)
  motor.motorVoltage[1] = velRef;
}
```

- 7) Implement the P-controller on the robot (for both motors individually)

- 8) Run with sample time 1ms (for debug)
- 9) Test with the same sample times as in MATLAB

- 10) Are the effects the same? We discuss and compare results next time.

```
%% sampling time
Ts = 0.001;
%
model = 'basebot_model_sampling';
aaa = sim(model, 0.6);
h = figure(171);
plot(aaa.velP.Time, aaa.velP.Data(:,1), LineWidth=2);
hold on
plot(aaa.velP.Time, aaa.velP.Data(:,2),
plot(aaa.velP.Time, aaa.velP.Data(:,3),
legend('ref (m/s)', 'Motor voltage/10 (V)')
title(sprintf('Sampling: Ts=%.3f sec, Kp=0.050'), Ts)
xlabel('Time (sec)')
ylabel('Velocity (m/s)')
grid on
saveas(h, 'sampling_time_1ms_step.png')
```



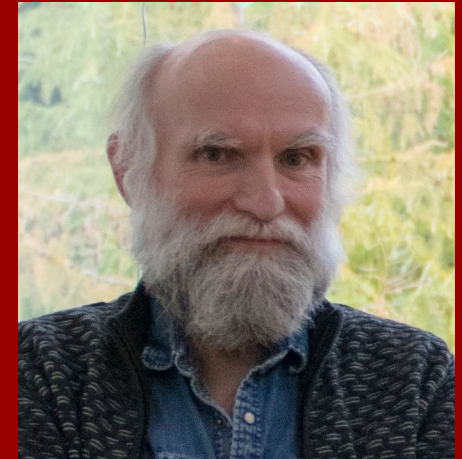
Issues?

211%

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Sampling effects

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Week 5